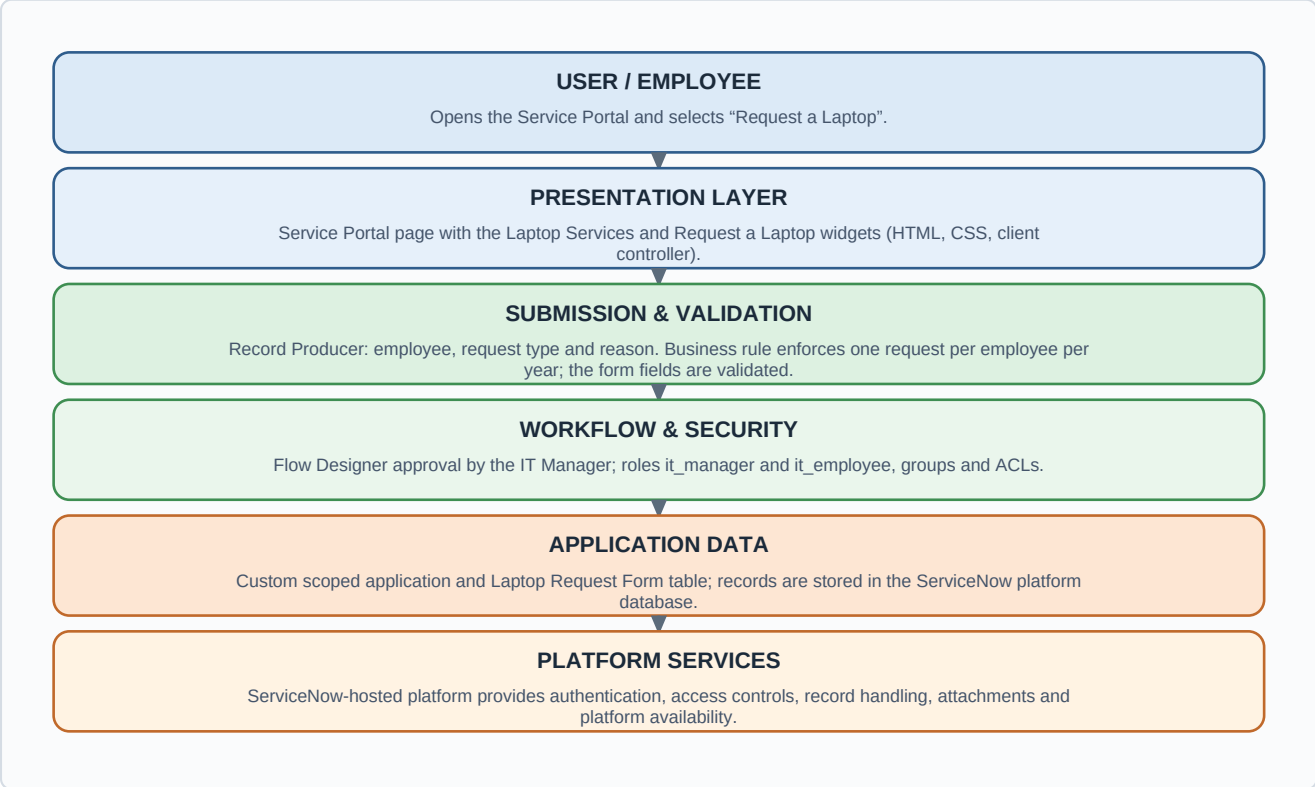
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	13 October 2026 (documentation date)
Team ID	N/A — Individual Project
Project Name	ServiceNow-Based Employee Laptop Request Application
Maximum Marks	4 Marks

Technical Architecture

The Employee Laptop Request Application is implemented on the ServiceNow platform. Employees access a Service Portal page and use a Record Producer to submit a laptop request. The submitted information is mapped to a custom Laptop Request Form table for validation, approval and tracking.



Architecture notes: No separate external API, machine-learning model, or independently configured cloud database is implemented in this project. The application uses the data storage and hosting provided by ServiceNow.

Table-1: Components & Technologies

No.	Component	Description	Technology
1	User Interface	Employee-facing portal page and laptop request form.	ServiceNow Service Portal; HTML, CSS, JavaScript/AngularJS widget controller
2	Application Logic-1	Captures and submits laptop request details through the Record Producer.	ServiceNow Record Producer and catalog variables
3	Application Logic-2	Blocks a second laptop request by the same employee in the same year and keeps the Number read-only.	ServiceNow business rule and field configuration
4	Application Logic-3	Routes each request to the IT Manager and updates the status to Approved or Rejected.	ServiceNow Flow Designer approval flow
5	Database	Stores submitted laptop requests and their fields.	Custom ServiceNow table: Laptop Request Form (u_laptop_request_form); ServiceNow platform database
6	Cloud Database	Database service is managed within the ServiceNow platform; no separate cloud database is configured.	ServiceNow-managed platform storage

No.	Component	Description	Technology
7	File Storage	Allows supporting files to be attached to a request, where enabled.	ServiceNow attachment functionality
8	External API-1	No external API is implemented.	Not applicable
9	External API-2	No second external API is implemented.	Not applicable
10	Machine Learning Model	No machine-learning model is part of the current implementation.	Not applicable
11	Infrastructure (Server / Cloud)	The application is developed and hosted on a ServiceNow developer instance and uses the platform's hosting services.	ServiceNow cloud platform; instance URL used for development and testing

Table-2: Application Characteristics

No.	Characteristic	Description	Technology
1	Open-Source Frameworks	The widget uses HTML, CSS and client-side JavaScript in the Service Portal. No separate open-source application framework is added.	HTML, CSS, JavaScript; ServiceNow Service Portal widget framework
2	Security Implementations	Access is handled through ServiceNow authentication and role-based access (roles, groups and ACLs). No custom encryption algorithm or firewall configuration is implemented.	ServiceNow platform authentication, roles and ACLs
3	Scalable Architecture	The solution uses a platform-hosted application and a custom table, so records are managed within ServiceNow. No separate microservices architecture is built.	ServiceNow platform architecture
4	Availability	The application relies on the availability of the ServiceNow instance and its platform services. No custom load balancer or distributed server setup is configured.	ServiceNow-hosted instance
5	Performance	The form uses catalog variables, a business rule and table-level choice configuration to standardize submission. No custom caching, CDN or throughput benchmark is implemented.	ServiceNow platform and native form, flow and widget behavior

References

• ServiceNow developer instance used to build and test the project.

• ServiceNow Service Portal, Record Producer and Flow Designer configuration used in the implementation.

• Project implementation: Laptop Request Form custom table, business rule, Record Producer, approval flow, roles and ACLs, and Service Portal widgets.

Note: This document describes the planned and implemented features of the project. Confirm the date and the exact table technical name against your instance before final submission.